

TG-40 (Annual QA)

Wednesday, July 31, 2024 7:46 AM



TG40 TG142
TG198...

★ Annual QA:

Subset of the tests performed during acceptance and commissioning

Baselines established, reconfirmed, or updated

Performed by QMP

PCAM standards used:

- Published department data
- AAPM TG-40/142
- Original Acceptance Test Data
- Varian Acceptance Test Specifications
- PA State Regulations

Annual Mechanical/Safety:

TABLE III. Annual.				TG-40	
Procedure	Machine-type tolerance				
	Non-IMRT	IMRT	SRS/SBRT		
Mechanical				2 mm diameter	Same Same Same Same 2 mm
Collimator rotation isocenter		±1 mm from baseline			
Gantry rotation isocenter		±1 mm from baseline			
Couch rotation isocenter		±1 mm from baseline			
Electron applicator interlocks		Functional			
Coincidence of radiation and mechanical isocenter	±2 mm from baseline	±2 mm from baseline	±1 mm from baseline		
Table top sag		2 mm from baseline		None Only vertical	Same Same Same Same
Table angle		1°			
Table travel maximum range movement in all directions		±2 mm			
Stereotactic accessories, lockouts, etc.	NA	NA	Functional		
Safety					
Follow manufacturer's test procedures		Functional			
Respiratory gating				None	N/A
Beam energy constancy		2%			
Temporal accuracy of phase/amplitude gate on		100 ms of expected			
Calibration of surrogate for respiratory phase/amplitude		100 ms of expected			
Interlock testing		Functional			

Annual Dosimetry:

TABLE III. Annual.				TG-40	
Procedure	Machine-type tolerance				
	Non-IMRT	IMRT	SRS/SBRT		
Dosimetry					
X-ray flatness change from baseline		1%		N/A	<u>1%</u> <u>2% Abs</u> <u>2%</u> <u>3% Abs</u>
X-ray symmetry change from baseline		±1%			
Electron flatness change from baseline		1%			
Electron symmetry change from baseline		±1%			
SRS arc rotation mode (range: 0.5–10 MU/deg)	NA	NA	Monitor units set vs delivered: 1.0 MU or 2% (whichever is greater) Gantry arc set vs delivered: 1.0° or 2% (whichever is greater)		
X-ray/electron output calibration (TG-51)		±1% (absolute)		<u>2%</u> <u>2%</u>	<u>2%</u> <u>1%</u>
Spot check of field size dependent output factors for x ray (two or more FSs)		2% for field size <4 × 4 cm ² , 1% ≥4 × 4 cm ²			
Output factors for electron applicators (spot check of one applicator/energy)		±2% from baseline			<u>2%</u>
X-ray beam quality (PDD ₁₀ or TMR ₁₀ ²⁰)		±1% from baseline			<u>PDD₁₀ 1%, other 2%</u> <u>R50 1mm, other 2mm</u> <u>None</u>
Electron beam quality (R ₅₀)		±1 mm			
Physical wedge transmission factor constancy		±2%			
X-ray monitor unit linearity (output constancy)	±2% ≥5 MU	±5% (2–4 MU), ±2% ≥5 MU	±5% (2–4 MU), ±2% ≥5 MU	<u>1%</u>	<u>1%</u>
Electron monitor unit linearity (output constancy)		±2% ≥5 MU			
X-ray output constancy vs dose rate		±2% from baseline		<u>N/A</u>	<u>2%</u>
X-ray output constancy vs gantry angle		±1% from baseline		<u>2%</u>	<u>1%</u>

Electron output constancy vs gantry angle	±1% from baseline	2%	1%
Electron and x-ray off-axis factor constancy vs gantry angle	±1% from baseline	2%	1%
Arc mode (expected MU, degrees)	±1% from baseline	None	Same
TBI/TSET mode	Functional	↓	Same
PDD or TMR and OAF constancy	1% (TBI) or 1 mm PDD shift (TSET) from baseline		Same
TBI/TSET output calibration	2% from baseline		Same
TBI/TSET accessories	2% from baseline		Same

Annual Imaging:

TABLE VI. Imaging.

			
	Application-type tolerance		
Procedure	non-SRS/SBRT	SRS/SBRT	
Annual (A)			Radiation Safety
Planar MV imaging (EPID)			
Full range of travel SDD	±5 mm	±5 mm	
Imaging dose ^e	Baseline	Baseline	
Planar kV imaging			
Beam quality/energy	Baseline	Baseline	
Imaging dose	Baseline	Baseline	
Cone-beam CT (kV and MV)			
Imaging dose	Baseline	Baseline	

*Annual review of imaging dose deemed sufficient due to minimal effect on overall dose and existing daily/monthly reviews of parameters that could affect dose

**UPenn Radiation Safety responsible for tube and exposure calibration checks

Annual VMAT guidelines (TG-218):

- Ensures linac can deliver VMAT plans that pass institutional VMAT QA criteria based on TG218 guidelines.
- Tests accuracy of VMAT delivery when treatment is interrupted and restarted

VMAT				
Test patient-specific VMAT QA constancy	Plan passes QA at a pass rate ≥ baseline pass rate minus 2% AND Plan passes institutional VMAT QA criteria	VMAT QA phantom	20–30 min	QMP
Interruption test	± 1% deviation from uninterrupted passing rate	VMAT QA phantom	10–15 min	QMP

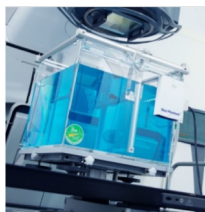
Annual Equipment:

► Blue Phantom 2 by IBA Dosimetry

- Large 3D water tank with robotic arms

Water Tank

Exterior water tank dimensions (LxWxH):	675 mm x 645 mm x 560 mm
Scanning volume (LxWxH):	480 mm x 480 mm x 410 mm
Position resolution:	0.1 mm
Position accuracy:	± 0.1 mm
Position reproducibility:	± 0.1 mm
Scanning speed:	max 50 mm/s
Approximate volume:	200 l
Wall thickness / material:	15 mm / acrylic
Weight:	45 kg



► 1D Scanner Water Tank by Sun Nuclear

- Smaller 1D scanner for output factors/TG-51



► Radiochromic Film

- Star Patterns to test isocentricity





Annual MLC/QA:

TABLE V. Multileaf collimation (with differentiation of IMRT vs non-IMRT machines).

Procedure	Tolerance	
	Weekly (IMRT machines)	
Qualitative test (i.e., matched segments, aka "picket fence")		Visual inspection for discernable deviations such as an increase in interleaf transmission
	Monthly	
Setting vs radiation field for two patterns (non-IMRT)		2 mm
Backup diaphragm settings (Elekta only)		2 mm
Travel speed (IMRT)		Loss of leaf speed >0.5 cm/s
Leaf position accuracy (IMRT)		1 mm for leaf positions of an IMRT field for four cardinal gantry angles. (<i>Picket fence</i> test may be used, test depends on clinical planning-segment size)
	Annually	
MLC transmission (average of leaf and interleaf transmission), all energies		±0.5% from baseline
Leaf position repeatability		±1.0 mm
MLC spoke shot		≤1.0 mm radius
Coincidence of light field and x-ray field (all energies)		±2.0 mm
Segmental IMRT (step and shoot) test		<0.35 cm max. error RMS, 95% of error counts <0.35 cm
Moving window IMRT (four cardinal gantry angles)		<0.35 cm max. error RMS, 95% of error counts <0.35 cm



[Picket fence](#)
[G0, 90, 180 270](#)

[DLG 0.6mm,](#)
[trans 0.5%](#)

[Same](#)
[Same](#)
[Same](#)

Wedge QA:

TABLE IV. Dynamic/universal/virtual wedges.

Dynamic-including EDW (Varian), virtual (Siemens), universal (Elekta) wedge quality assurance				
Frequency	Procedure	Tolerance		
		Dynamic	Universal	Virtual
Daily	Morning check-out run for one angle		Functional	
Monthly	Wedge factor for all energies	C.A. axis 45° or 60° WF (within 2%) ^a	C.A. axis 45° or 60° WF (within 2%) ^a	5% from unity, otherwise 2%
Annual	Check of wedge angle for 60°, full field and spot check for intermediate angle, field size	Check of off-center ratios @ 80% field width @ 10 cm to be within 2%		

^aRecommendation to check 45° if angles other than 60° are used.



[60° \(2%\)](#)

[Off axis ratio](#)
[2%](#)